

Strategic Practice 2

1/6/25 - 1/8/25

I'm doing things differently:

1) using an iPad so my work is automatically digitized

↳ writing with a stylus is weird !!

2) I'm going to do PSETS from the STAT110 website since the questions are...

↳ fewer LOL

↳ still have solutions so I can check my work

↳ are real PSETS from the course ☺

Since I did questions from the book for Ch 1, I'm starting w/ PSET 2 and skipping the first section

2.1) A & B are independent events if $P(A \cap B) = P(A)P(B)$

If $P(A) > 0$ and $P(B) > 0$, then this is equivalent to $P(A|B) = P(A)$ and $P(B|A) = P(B)$

NOT QUIZE

An event can NOT be independent of itself because if $P(A) > 0$, then $P(A \cap A) = P(A) = P(A)^2$ and $P(A \cap A) \neq [P(A)]^2$. If $P(A) = 0$, the event is still NOT independent of itself b/c $P(A|A) = 1 \neq 0 = P(A) = 0$

this can be true when $P(A) = 0$ or 1

2.2) If A and B are independent, then A^c and B^c are independent too ✓

$$\begin{aligned} P(B^c|A^c) &= 1 - P(B|A^c) \\ &= 1 - \frac{P(A^c|B)P(B)}{P(A^c)} \end{aligned}$$

Alternate proof

$$\begin{aligned} P(A^c \cap B^c) &= 1 - P(A \cup B) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] \end{aligned}$$

$$\begin{aligned}
 P(A^c) &= 1 - P(A) - P(B) - P(A)P(B) \\
 &= 1 - \frac{(1 - P(A|B))P(B)}{P(A^c)} = (1 - P(A))(1 - P(B)) \\
 &= 1 - \frac{P(A^c)P(B)}{P(A^c)} = P(B^c)
 \end{aligned}$$

2.3) Example where events A, B, and C are pairwise independent but NOT independent

Flip = coin (fair coin) twice

A: event first coin shows H

B: event second coin shows T

C: event coin flips are either HT or TH

$$P(A \cap B) = \frac{1}{4} = P(A)P(B) \checkmark$$

$$P(A \cap C) = \frac{1}{4} = P(A)P(C) \checkmark$$

$$P(B \cap C) = \frac{1}{4} = P(B)P(C) \checkmark$$

$$\text{However, } P(C | A, B) = 1 \neq \frac{1}{2} = P(C)$$

2.4) 3 events that are NOT independent and satisfy $P(A \cap B \cap C) = P(A)P(B)P(C)$

roll = 6-sided die twice

A: event you roll a 7

B: event you roll a 1

C: event you roll a 1 twice

B & C are clearly NOT independent since B would provide information about C.

However, since $P(A) = 0$,

$$P(A \cap B \cap C) = P(A)P(B)P(C) = 0$$

$$\begin{aligned}
 3.1) & P(\text{second marble green} | \text{first marble green}) \\
 &= \frac{P(\text{second marble green} \cap \text{first marble green})}{P(\text{first marble green})} \\
 &= \frac{0.5}{1 \times 0.5 + 0.5 \times 0.5} = \frac{2}{3}
 \end{aligned}$$


 this is
Bayes's!

previous method
 which FWIW I think
 is valid!

original marble	second marble
0.5 blue	green
0.5 green	green

$$\begin{aligned}
 & \xrightarrow{\text{both green}} \\
 & P(\text{second marble green} | \text{first marble green}) \\
 &= \frac{P(\text{first marble green} \cap \text{second marble green})}{P(\text{first marble green})} \\
 &= \frac{1 \cdot \frac{1}{2}}{\frac{3}{2}} = \frac{2}{3}
 \end{aligned}$$

- 3.2) 80% of email is spam
 10% of spam emails have/contain "free money"
 1% of non-spam emails contain "free money"

$$\begin{aligned}
 & P(\text{email is spam} | \text{email contains "free money"}) \\
 &= \frac{P(\text{email is spam} \cap \text{email contains "free money"})}{P(\text{email contains "free money"})} \\
 &= \frac{0.80 \times 0.10}{0.80 \times 0.10 + 0.20 \times 0.01} = \frac{40}{41}
 \end{aligned}$$


 this is Bayes's!

- 3.3) G : event that a certain individual is guilty
 of a robbery

we learn an event E_1 occurred and then a
 while later an event E_2 occurred

- 2) yes, it's possible that $P(G|E_1) > P(G)$ and
 $P(G|E_2) > P(G)$ and $P(G|E_1, E_2) < P(G)$

Here's an example:

(like the solution's example story).

*Robbery committed b/w
1-3 PM.
A: event suspect was at a shop nearby b/w 1-2 P.M.
B: " " " " " "
C: suspect is guilty*

b) computing $P(G | E_1, E_2)$ in one step

OK

$$2) P_{\text{new}}(G|E_2) = \frac{P_{\text{new}}(E_2|G) P_{\text{new}}(G)}{P_{\text{new}}(E_2)}$$

$$= P(G|E_1, E_2) \text{ conditioning on } E_1$$

Crime is committed by one of them

Equal evidence against both of them

$$\hookrightarrow P(A \text{ is guilty}) = P(B \text{ is guilty})$$

0.5

0.5

Further investigation found that the guilty party has a blood type found in 10% of the population

A matches this blood type whereas B's blood type match is unknown.

a) given new info what is the probability that A is guilty?

$$P(G_A | M_A) = \frac{P(M_A | G_A) P(G_A)}{P(M_A)} = \frac{0.5}{1.1 \times 0.5} = \frac{10}{11}$$

$P(M_A | G_A) = 1$ if suspect A committed the crime, then suspect A's blood type must match the guilty party's

$$P(G_A) = 0.5$$

$$P(M_A) = P(M_A | G_A) P(G_A) + P(M_A | G_A^c) P(G_A^c) \\ = 1 \times 0.5 + 0.1 \times 0.5$$

LOTP!

b) Given this new information, what is the probability that B's blood type matches that found at the crime scene

M_B = event B's blood type matches that at the crime scene

LOTP & conditioning

$$P(M_B | M_A) = P(M_B | G_A, M_A) P(G_A | M_A) \\ + P(M_B | G_A^c, M_A) P(G_A^c | M_A) \\ = 0.1 \times \frac{10}{11} + 1 \times \frac{1}{11} = \frac{2}{11}$$

3.5) you are going to play 2 games of chess

your opponent is equally likely to be a beginner, intermediate, or master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30% respectively

a) $P(\text{winning first game})?$

F : event of winning first game

$$\begin{aligned} P(F) &= P(F|\text{beginner})P(\text{beginner}) + \\ &\quad P(F|\text{intermediate})P(\text{intermediate}) + \\ &\quad P(F|\text{master})P(\text{master}) \\ &= \frac{1}{3} \left(\frac{3}{10} + \frac{5}{10} + \frac{9}{10} \right) = \frac{17}{30} \end{aligned}$$

b) S : event of winning the second game

$$P(S|F) = \frac{P(F|S)P(S)}{P(F)} \quad \rightarrow \quad \begin{array}{l} \text{dead end b/c} \\ \text{we don't know} \\ P(F|S) \end{array}$$

Going to try LOTP

B : event you play beginner

I : event you play intermediate

M : event you play master

$$\begin{aligned} P(S|F) &= P(S|B, F)P(B|F) + \\ &\quad P(S|I, F)P(I|F) + \\ &\quad P(S|M, F)P(M|F) \\ &= 0.9 \times P(B \cap F)/P(F) + \\ &\quad + 0.5 \times P(I \cap F)/P(F) + \\ &\quad + 0.3 \times P(M \cap F)/P(F) \\ &= 0.9 \times \frac{9}{17} \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{b/c of} \\ \text{assumption} \\ \text{that given} \\ \text{skill level} \\ \text{of opponent,} \\ \text{outcome of} \end{array}$$

$$\begin{aligned}
 & + 0.5 \times \frac{5}{17} \\
 & + 0.3 \times \frac{3}{17} \\
 & = \frac{1}{17} (8.1 + 2.5 + 0.9) = \frac{11.5}{17} = \frac{23}{34}
 \end{aligned}$$

games are independent

$$P(B|F) = P(F|B) \frac{P(B)}{P(F)} = 0.90 \frac{\frac{1}{3}}{\frac{17}{30}} = \frac{9}{10} \cdot \frac{30}{17} \cdot \frac{1}{3}$$

$$P(I|F) = P(F|I) \frac{P(I)}{P(F)} = 0.50 \frac{\frac{1}{3}}{\frac{17}{30}} = \frac{5}{10} \cdot \frac{30}{17} \cdot \frac{1}{3}$$

$$P(M|F) = P(F|M) \frac{P(M)}{P(F)} = 0.30 \frac{\frac{1}{3}}{\frac{17}{30}} = \frac{3}{10} \cdot \frac{30}{17} \cdot \frac{1}{3}$$

- c)

Assumption 1 outcomes of the games are independent	Assumption 2 outcomes of games are conditionally independent given opponents skill level
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Assumption 1 says that the outcome of game 1 gives no information about the outcome of game 2. Assumption 2 says that given the skill-level of your opponent, the outcome of each game gives no information about the outcome of the next.

Assumption 2 is more reasonable. It's hard to see how the outcome of game 1 would impact the outcome of game 2 if you are still playing the same opponent. Their skill level is unlikely to meaningfully change & yours is unlikely to either.

Assumption 1 is less reasonable. If you win game 1, it gives you info that you are more likely playing a beginner than the intermediate or advanced player.